Chapter 3 Answers

The proofs for the propositions 1.1-1.2 actually prove the corresponding propositions for the other three sections. Hence, we only prove the theorems in section 1, while the others can be easily seen to follow from these proofs.

1.1

This is a consequence of 1.5 and 1.6, as 1.6 shows that any universal register is creative, while 1.5 shows that any creative register can be stumped (and hence any universal register can be stumped).

1.2

Suppose that at least one of the registers, call it R_n, is universal. Hence, for all numbers x and y, R_n affirms xy iff R_x affirms y. Now, $R_{n^{\#'}}$ affirms $n^{\#'}$ iff R_n affirms $n^{\#'}n^{\#'}$ iff $R_{n^{\#}}$ affirms $n^{\#'}$ iff $R_{n^{\#'}}$ denies $n^{\#'}$. Thus, $R_{n^{\#'}}$ affirms $n^{\#'}$ iff $R_{n^{\#'}}$ defines $n^{\#'}$, thus meaning $R_{n^{\#'}}$ is stumped by $n^{\#'}$, its own index.

1.3

This, like 1.1, is a consequence of 1.5 and 1.6.

1.4

(a): Suppose some register R_n is universal. Thus $R_{n^{\#}}$ affirms $n^{\#'}$ iff R_n affirms $n^{\#'}n^{\#'}$ iff $R_{n^{\#'}}$ affirms $n^{\#'}$ iff $R_{n^{\#}}$ denies $n^{\#'}$, meaning $R_{n^{\#}}$ is stumped

by $n^{\#'}$, the index of its opposer.

(b): Next, suppose that some register R_n is contra-universal. Hence, for all x and y, $R_{n'}$ affirms xy iff R_n denies xy iff R_x affirms y. Thus $R_{n'}$ is a universal register. Thus, because there is a universal register, we know that some register is stumped by its own index by Proposition 1.2. In particular: $R_{n^\#}$ affirms $n^\#$ iff R_n affirms $n^\# n^\#$ iff $R_{n^\#}$ denies $n^\#$, thus proving that $R_{n^\#}$ is stumped by $n^\#$, its own index.

1.5

Suppose some register C is creative. Thus for every register R, there is at least one number n such that C affirms n iff R affirms n. Well, consider $R_{k'}$ for k the index of C. Thus there is at least one number n such that R_k (that is, C) affirms n iff $R_{k'}$ affirms n iff R_k denies n, and hence R_k (C) must be stumped by n.

1.6

(a): First, we show that every universal register is creative. Suppose U is universal, that is, for all numbers x and y, U affirms xy iff R_x affirms y. Now, consider any register R_n. We must show that there is at least one number m such that U affirms m iff R_n affirms n. Well, U affirms $n^\# n^\#$ iff $R_{n^\#}$ affirms $n^\#$ iff R_n affirms $n^\# n^\#$, thus proving that U is creative.

(b): Next, we must show that every contra-universal register is creative. Suppose V is contra-universal, that is, for all numbers x and y, V affirms xy iff R_x denies y. Now, consider any register R_n. We must show that there is at least one number m such that U affirms m iff R_n affirms m. Well, V affirms $n^{\#'} n^{\#'}$ iff $R_{n^{\#'}}$ denies $n^{\#'}$ iff $R_{n^\#}$ affirms $n^{\#'}$ iff R_n affirms $n^{\#'} n^{\#'}$, thus proving that V is creative.

1.7

Consider some register R_x. Now, R_x affirms $x^\# x^\#$ iff $R_{x^\#}$ affirms $x^\#$ (which is to say $x^\# x^\# \in A$) and R_x denies $x^\# x^\#$ iff $R_{x^\#}$ denies $x^\#$ (which is to say $x^\# x^\# \in B$). Hence for every register R_x, $x^\# x^\#$ is a fixed point of R_x.

1.8

We suppose that U is universal and V is contra-universal.

(a): Suppose xy is a fixed point of U'. Hence, for every x, R_x affirms y iff U affirms xy iff U' denies xy iff $xy \in B$ iff R_x denies y, thus meaning U is stumped by xy, the fixed point of U'.

(b): Suppose xy is a fixed point of V. Hence R_x denies y iff V affirms xy iff $xy \in A$ iff R_x affirms y, thus meaning V is stumped by xy, its own fixed point.

1.9

Suppose some register R is such that R affirms all number in A and doesn't affirm any number in B. Hence, if $xy \in A$, then R affirms xy, while if $xy \in B$, A doesn't affirm xy (i.e. either denies or is stumped by it). Let $R_k = R$ (that is, let k be the index of R).

(a): On the one hand, let $k^{\#'}k^{\#'}$ be in A. Then $R_{k^{\#'}}$ would affirm $k^{\#'}$ and R_k would affirm $k^{\#'}k^{\#'}$, and thus $R_{k^{\#}}$ would deny $k^{\#'}$ and thus R_k would deny $k^{\#'}k^{\#'}$, which is a contradiction. Thus $k^{\#'}k^{\#'} \notin A$.

(b): On the other hand, suppose that $k^{\#'}k^{\#'} \in B$. Then $R_{k^{\#'}}$ would deny $k^{\#'}$ and R_k would not affirm $k^{\#'}$, and thus $R_{k^{\#}}$ would affirm $k^{\#'}$ and thus R_k would affirm $k^{\#'}k^{\#'}$, which is a contradiction. Thus $k^{\#'}k^{\#'} \notin B$.

Hence $R_{k^{\#'}}$, because $k^{\#'}k^{\#'}$ is in neither A nor B, must be stumped by $k^{\#'}$. So R_k affirms $k^{\#'}k^{\#'}$ iff $R_{k'}$ denies $k^{\#'}k^{\#'}$ iff $R_{k^{\#'}}$ affirms $k^{\#'}k^{\#'}$, meaning R_k can neither affirm nor deny $k^{\#'}k^{\#'}$, and hence is stumped by it.

Note: Smullyan said in his answer that we could show that R_k is stumped by xy for xy the fixed point of R. However, I was unable to see this.

1.10

Suppose that some register R_k is a stump detector. Thence R_k halts for $k^{\#}k^{\#}$ iff $R_{k^{\#}}$ is stumped by $k^{\#}$ iff R_k is stumped by $k^{\#}k^{\#}$. Thus R_k halts for $k^{\#}k^{\#}$ iff R_k is stumped by $k^{\#}k^{\#}$, thus meaning R_k cannot exist.

1.11

We suppose that U is universal, V is contra-universal, and that R dominates U and R' dominatnes V. Now, every number xy in A is such that R_x affirms y, meaning U affirms xy and hence R affirms xy. Also, any number xy in B is such that R_x denies y meaning V affirms xy meaning R' affirms xy and thus R denies xy. Hence R affirms all numbers in A and doesn't affirm any number in B. Thus, by Proposition 1.9, R can be stumped.

1.12

We suppose that there is some universal register U. Now, let S be the affirmation set for U. Hence, because, for all x and y, U affirms xy iff R_x affirms y, S contains all those and only those numbers xy such that R_x affirms y (hence S is the set A). Hence S' (that is, the complement of S) contains all those and only those numbers xy such that R_x does not affirm y. But suppose that there was some register R such that S' was R's affirmation set. Now, this would mean that R would affirm all those and only those numbers xy such that R_x does not affirm y. Let k be the index of R. Hence R_k affirms $k^{\#}k^{\#}$ iff $R_{k^{\#}}$ did not affirm $k^{\#}$ (i.e. either denies or is stumped by it) iff R_k did not affirm $k^{\#}k^{\#}$, which means such an R_k cannot exist. Therefore S' (that is, A') is not an affirmation set.

Alternatively: We can instead show there cannot be any register R_k (as in the above) using fixed points. Consider some R_k such that S' is its affirmation set. Let xy be its fixed point. Thus R_k affirms xy iff $xy \in A$ and R_k denies xy iff $xy \in B$. So, if $xy \in A$, R_k affirms A, while if $xy \in A'$, R_k does not affirm xy. In any case, S' (that is, A') cannot be the affirmation set for any register R_k, as either R_k affirms some number xy outside of A', or A' contains some number xy such that R_k does not affirm xy (both of these cases would mean S' and the set of numbers affirmed by R_k are not the same).